through the Vacuum pipe. The control rack connected with the diaphragm is operated and the diesel supply is controlled. By this way the engine speed is controlled in the pneumatic governor.

3. **Hydraulic Governor :** This type of governor is operated by the cam shaft of the engine. The control plunger in this governor is connected with the control rack of the fuel injection pump by a spring. This type of governors are operated by the liquid present in the control plunger. When engine is started due to the rotation of the cam shaft the control plunger is operated and the control plunger operates the control rack of the fuel injection pump. The control rack of the fuel injection pump is operated by the hydraulic liquid in the control plunger. The schematic view of the hydraulic governor shown in Fig 9.10.1.3.

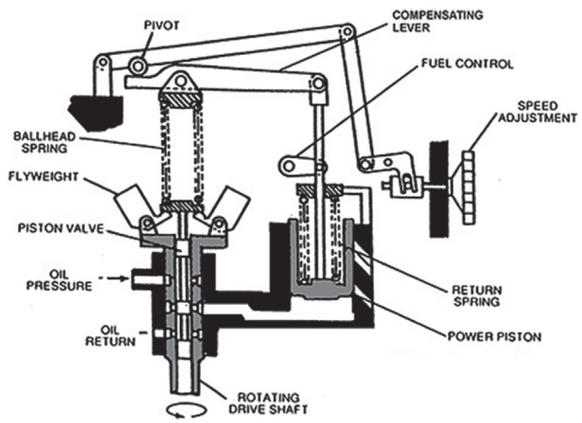


Fig 9.10.1.3 Schematic View of the Hydraulic Governor

9.11 DIESEL KNOCK

Knock in diesel engine is an unwanted effect caused due to the increased ignition delay of the engine. During the ignition delay period fuel is injected continuously. When the accumulation of the fuel is more due to the increased ignition delay, when ignited the accumulated fuel suddenly gets combusted and results in severe fluctuation in cylinder pressure and rate of pressure rise. This cause violent sound called as diesel knock.

9.11.1 *Reasons for Diesel Knock*

1. Lower compression ratio of the engine
2. Lower fuel injection pressure
3. Faulty injector
4. Blockage in the nozzle
5. Earlier injection of the diesel fuel

9.12 COMMON RAIL DIRECT INJECTION

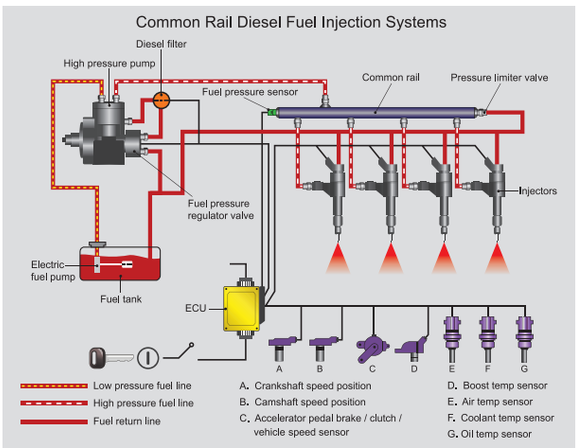


Fig.9.12 Common Rail Direct Injection

Common Rail direct injection of diesel fuel is the recent technology followed in modern diesel engines in introducing diesel fuel inside the engine combustion chamber. To improve the vaporization behaviour of the diesel fuel, the fuel is supplied to a common rail at very high pressure from the fuel pump. This method is called as the common rail direct injection. In this method with a small amount of diesel fuel more power is obtained. In addition the emission levels are reduced

AM_Unit_09.indd 194 31-01-2019 18:02:30